APPENDIX

2

Familial DNA Searches: Potential, Pitfalls, and Privacy Concerns

Privacy rights groups feel that familial searching is going too far. In March 2008, the FBI Laboratory organized a meeting held in Washington, D.C. to discuss the advantages and concerns with familial searching. Scientists, lawyers, and legal scholars presented their points of view (see Rozen 2009). Although no formal proceedings were published from this meeting, the forum permitted a vigorous discussion of the issues by proponents and opponents of familial searching.

Since a successful California familial search was announced in July 2010 (D.N.A. Box 8.5, Miller 2010), a number of laboratories and lawmakers are considering the potential of familial searching. However, the technique has technical pitfalls and privacy concerns that will be discussed below.

As of 2010, two states within the U.S. permit active familial searching (California and Colorado), two ban familial searching (Maryland and the District of Columbia), and several others are exploring the possibility of this extended searching (Virginia) or to allow follow up on partial matches (New York).

An examination of the differences between a direct match, a partial match, and no match is helpful to begin our discussion of familial searching, how it works, and its limitations. In Figure A2.1, three different suspect profiles are compared against an evidentiary DNA profile. Suspect 1 matches at all 10 alleles shown—and is therefore a complete or direct match. Suspect 3 does not match the evidence (is excluded) as the profiles fail to share any alleles at two of the loci and only has a single allele in common at the remaining three loci. However, Suspect 2 is a partial match or what may be termed a "near miss" because only 7 out of 10 possible alleles are in common between the compared samples. Note that some of the loci, such as D8S1179 and D13S1358, share two alleles while the remaining three loci only have a single allele in common at each locus. In terms of the CODIS software match stringency discussed in Chapter 8, these single allele-sharing loci exhibit low stringency (L) rather than

(a)

	D8S1179		D21S11		D7S820		CSF1PO		D3S1358		
Evidence	**13**	**14**	**31**	**31.2**	**9**	**11**	**12**	**13**	**15***	**15**	
Suspect 1	**13**	**14**	**31**	**31.2**	**9**	**11**	**12**	**13**	**15***	**15**	**Direct Match**
Alleles shared	*2*		*2*		*2*		*2*		*2*		*10 alleles shared*
Match stringency	H		H		H		H		H		

(b)

Evidence	**13**	**14**	**31**	**31.2**	**9**	**11**	**12**	**13**	**15**	**15**	
Suspect 2	**13**	**14**	30	**31.2**	**9**	10	9	**12**	**15**	**15**	**Partial Match**
Alleles shared	*2*		*1*		*1*		*1*		*2*		*7 alleles shared*
Match stringency	H		L		L		L		H		

(c)

Evidence	**13**	**14**	**31**	**31.2**	**9**	**11**	**12**	**13**	**15**	**15**	
Suspect 3	12	**13**	29	30	10	**11**	**12**	12	14	14	**No Match**
Alleles shared	*1*		***0***		*1*		*1*		***0***		*3 alleles shared*
Match stringency	L		--		L		L		--		**Excluded at 2 loci**

*Although shown as two alleles (“15,15”) in this example, homozygous genotypes are recorded as only a single allele within the CODIS software (i.e., “15” instead of “15,15”).

FIGURE A2.1 Three different suspect profiles are compared against an evidentiary DNA profile (only 5 of 15 tested STR loci are shown here for space purposes). (a) Suspect 1 matches at all 10 alleles. (b) Suspect 2 is a partial match or a “near miss” with 7 out of 10 possible alleles being shared. Note that at least one allele is shared with the evidentiary profile at every locus suggesting that there is a possible parent-child relationship. (c) Suspect 3 does not match the evidence because he is excluded at 2 loci and only shares 3 out of 10 possible alleles.

high stringency (H) where both alleles match between the compared samples. (Note that routine CODIS searches are done at moderate stringency, and low stringency searching is only used in missing persons cases when a parent is looking for a missing child or vice versa.)

Alleles can be shared between samples due to common ancestry (identical by descent, IBD) or have the same characteristics (e.g., repeat number) by chance (identical by state, IBS). Alleles that are more common (i.e., have a higher frequency) will have a greater chance of being IBS. When rare alleles are present in a DNA profile, there is a greater chance of them being IBD with a matching profile and thus useful in familial searching.

There are two primary methods of expressing the strength of a match between samples. The first is allele sharing, which simply tabulates the number of alleles in common. The second is the likelihood ratio (LR), which compares the hypothesis that the samples being compared are related versus the hypothesis that they are unrelated. LRs will be discussed in greater detail in the forthcoming volume *Advanced Topics in Forensic DNA Typing: Interpretation*.

For a father-child comparison, the LR is known as the paternity index (PI). For sibling comparisons, the LR is referred to as the sibship index (SI). When multiple loci are considered, a combined paternity index (CPI) or combined sibship index (CSI) is calculated by multiplying the individual PI or SI values for each locus. More generally, either the PI or SI can be called a kinship index (KI). The LR calculation is more informative than simply counting the number of alleles shared because it incorporates the frequencies for those alleles observed in a profile. Sharing rare alleles is less likely with unrelated individuals and therefore will create a higher LR value.

Even though relatives share more DNA than unrelated individuals, there will be partial matches with unrelated individuals especially if the crime scene sample contains common alleles. These common alleles can give rise to false positives in a familial search.

Pitfalls of False Positives and False Negatives

The SWGDAM recommendations on partial matches note that current CODIS searching rules and algorithms, which use allele sharing (and thus cannot incorporate information from rare alleles) have a very low efficiency for locating true relatives in offender databases (SWGDAM 2009). These recommendations further comment that true siblings very rarely share alleles at all 13 CODIS loci and the number of unrelated individuals who do share at least one allele at all loci increases as offender DNA databases get larger. Thus, LR methods that enable allele frequency as well as allele sharing information are typically preferred for familial searching.

Tom Reid and colleagues at DNA Diagnostics Center, a paternity testing laboratory near Cincinnati, Ohio, studied the ability of different search strategies to locate true siblings within a simulated offender database (Reid et al. 2008). One of each pair from 109 sibling pairs with CSI values ranging from four to greater than one billion were added to a simulated offender database containing 12,292 profiles. The remaining sibling from each pair was used to probe the database to see if its partner (familial) profile could be located within the first 100 best matches in the database. An allele sharing method detected 62 of the 109 known sibling pairs (57%) while a likelihood ratio (kinship matching) approach detected 90 of the sib pairs (83%).

Another study using a different set of STR loci found a 72% chance of the true sibling being in the top 100 profiles using allele sharing and a 78% chance using a likelihood ratio approach that takes into account the rarity of the alleles being shared (Curran & Buckleton 2008). It is worth noting that these simulated databases of a few thousand DNA profiles are much smaller than a state DNA database that contains hundreds of thousands or even

greater than a million profiles like California. It is likely that these published approximately 70% to 80% chances will dramatically decrease with increased database size.

Not surprisingly, the Reid et al. study found that the efficiency of a search (how high up the ranked list the true sibling was) was inversely correlated with the CSI value. The presence of rarer alleles in the profile being shared between siblings increases the CSI value and hence the probability of matching.

Of the sibling pairs having a CSI value greater than 10,000, all but one pair matched on the first hit (i.e., were first on the rank-ordered database match list). Sibling pairs with CSIs between 1000 and 10,000 averaged approximately three false hits to obtain a correct match. However, sibling pairs with CSIs between 10 and 100 required an average of 52 false hits before a true match was found (Reid et al. 2008).

The DNA Diagnostics Center study demonstrates that sibling searches will not always work because not all loci share alleles as with parent-child matches (barring mutations). The authors conclude that without further data filtering using geography or other factors, "familial searching using DNA profile data may not consistently produce enough usable leads to make it meaningful" (Reid et al. 2008).

CSI values can vary within a single family due to allele inheritance. For example, I have four brothers whose 13-locus CSI values (relative to being my brother) are 2300, 490, 1.5, and 0.83. While my first two brothers with likelihood ratios of 490 and 2300 might be identified in a familial search, it is highly unlikely that my lower-CSI-value brothers would. Many other non-related individuals would likely share alleles with me by chance and be higher on a ranked list than my true brothers. If my DNA profile was the crime scene profile being searched and if my brothers with low CSI values of 1.5 and 0.83 were in the database, our true familial connection would be missed and result in false negatives for that search. They are my brothers, they are in the database, but due to allele inheritance patterns where we inherited different non-overlapping alleles from our parents, a familial search based on the CODIS 13 autosomal STR loci would not identify them.

To filter out false positives or adventitious matches due to chance allele sharing (D.N.A. Box 8.3) following a familial search, further information is needed to narrow possibilities. Metadata, such as geographic information or possible ages of individuals if available, can be effective and has been used in the UK. Alternatively, additional loci, such as Y-STRs (Bieber et al. 2006, Myers et al. 2010), that are run on the crime scene sample and the list of potential familial offenders can eliminate male individuals with different paternal lineages.

Barring mutations, true fathers, sons, and full brothers will have the same Y-STR haplotype (see Chapter 13). If additional loci are typed on the putative familial offender profiles after the familial search, as long as the profiles of the potential relatives are in the ranked list for comparison purposes, the Y-STR testing should filter out most if not all of the non-relatives as occurred in the Grim Sleeper case (D.N.A. Box 8.5).

Low Success Rates

The UK, which has the most experience so far with attempting familial searches, has reported a success rate of anywhere between 11% and 27%. Chris Maguire, representing the Forensic Science Service at the 21st International Symposium on Human Identification meeting held in October 2010, reported 35 successful uses of familial searches on 131 cases

(≈27%) where the approach was attempted. However, an *LA Times* article on the topic published in November 2008 claimed that "British authorities" reported 158 familial searching attempts with only 18 successful leads (≈11%) (Dolan & Felch 2008). Regardless of the actual percentage, the success rate is low, particularly when only 10 STRs are being used (with the SGM Plus kit). This low success rate is due to the many false positives and false negatives that can arise when attempting to make kinship comparisons with 10 STRs or to the lack of a close relative on the database.

The Grim Sleeper case described in D.N.A. Box 8.5 was the tenth familial search attempted by California—and so far is the only successful familial search in this state (Myers et al. 2010). Thus, at this early stage in the California program the success rate is about 10%. Low success rates result from a lack of close relatives of the perpetrator in the database or from false negatives where true relatives in the database are missed due to the search strategy used and that only 13 or 15 autosomal STR loci are typically available in the initial search. More autosomal loci can increase the power to perform familial searches (O'Connor et al. 2010), but would have to be run on both the offender samples and the crime scene stain being used for the familial search.

Lineage Markers to Improve Success Rates

In order to reduce the number of false positives (because of sharing common alleles) and false negatives (because of genetic inheritance patterns in the family members), lineage markers are necessary to increase success rates. The use of Y-STRs and mtDNA in combination with autosomal results improves the success of familial searching (Lewis 2009). However, using these additional markers requires that both the evidentiary profile and the searched database samples have the same loci. False positives can be reduced by using Y-STR testing on male samples identified in a ranked list (Myers et al. 2010). Unfortunately, as noted above, false negatives will be missed unless the additional loci are tested on all or a significant portion of the database samples.

Supposed "Genetic Surveillance" of Families

In their coverage of the familial searching issue, reporters and law professors often raise the specter of so-called "genetic surveillance"—that families will come under some kind of oppressive suspicion because they are related to an offender whose DNA profile resides on a DNA database (NY Times 2010, Rosen 2009, Suter 2010, Murphy 2010). Loosening the search parameters does not in and of itself produce associations to family members. There is greater uncertainty in the results returned following a familial search because a direct match is not involved (Figure A2.2). The most significant problem with familial searching is the production of false positives due to the crime profile having common alleles that are shared by unrelated individuals, potentially producing likelihood ratio values indicative of a putative relationship. Even with relatively low LRs for individual loci, the combination of LRs across 13 to 15 STR loci can produce fairly large LRs that appear high in a ranked list.

As was done in the Grim Sleeper case, additional screening using Y-STR typing needs to be performed prior to sharing any information with investigators so that false leads are minimized or eliminated. Perhaps even more importantly, the California familial search policy states that no potential candidate names can be shared with the requesting agency

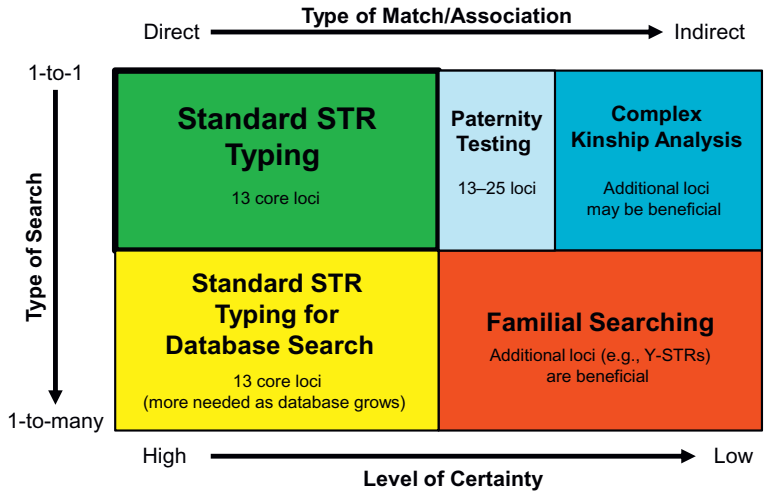


FIGURE A2.2 Illustration of the relationship between the types of matches being attempted, the types of searches being conducted, and the numbers of markers used. Familial searching attempts to identify a parent-offspring or full sibling relationship by comparing an evidence profile to the entire database of offender profiles. This one-to-many search will result in fortuitous matches due to chance allele sharing, and not allele sharing due to a common ancestor. Figure courtesy of Kristen Lewis O'Connor, NIST.

(i.e., Los Angeles Police Department in the Grim Sleeper case) until further investigation is performed using public records. If a candidate familial association is identified, investigative footwork through public records searches is performed. Within the California Department of Justice, investigators are careful not to talk to people (e.g., potential family members) and the requesting agency until the candidate familial association has cleared the investigative and scientific hurdles (Steinberger & Sims 2008). Thus, if everything is done right, no one (outside of the investigative team) knows when a candidate familial association does not pan out as a true relative of the perpetrator.

It is important to keep in mind that familial searches, as presently conducted, can produce many false negative results with true siblings and work best with father-son situations, where alleles are shared across an entire profile as in the Grim Sleeper case (D.N.A. Box 8.5). However, the majority of convicted offenders with DNA profiles in the database do not have children old enough to be committing crimes (see Table 8.10).

As of 2010, national databases have been in operation for about 15 years. As their time of use expands, the number of relatives in DNA databases is likely to increase as children of previous offenders could start to be included. In the Grim Sleeper case (D.N.A. Box 8.5), the son's newly added profile helped catch a father who was not in the database. Conversely with aging DNA databases and familial searching capabilities, a father's profile could lead to a son's apprehension as the younger man begins a life of crime.

The data in Table 8.10 illustrate that the largest numbers of individuals in prison are 20 to 45 years of age. This range is roughly the span of a generation or 25 years. Hence, where family members may be in prison (and therefore have their DNA profiles in databases enabling present-day familial searching), we might infer that siblings are more likely

than parent-offspring to be present. Siblings are less likely to share alleles with one another than parent-offspring. However, as databases are used for longer periods of time, they may begin to include the offspring of former offenders. If this occurs, then there will begin to be a higher number of relatives on the database with a greater chance of allele sharing. More genetic markers will be needed in DNA databases of the future to help discriminate between familial and random matches.

READING LIST AND INTERNET RESOURCES

Familial Searching

Bieber, F. R., et al. (2006). Finding criminals through DNA of their relatives. *Science*, *312*, 1315–1316.

Bieber, F. R. (2006). Turning base hits into earned runs: Improving the effectiveness of forensic DNA data bank programs. *Journal of Law, Medicine & Ethics*, *34*, 222–233.

Budowle, B. (2010). Familial searching: Extending the investigative lead potential of DNA typing. *Profiles in DNA*, *13*(2). Available at <http://www.promega.com/profiles/1302/1302_07.html>.

Chung, Y.-K., et al. (2010). Familial database search on two-person mixture. *Computational Statistics and Data Analysis*, *54*, 2046–2051.

Cowen, S., & Thompson, J. (2008). A likelihood ratio approach to familial searching of large DNA databases. *Forensic Science International: Genetics Supplement Series*, *1*, 643–645.

Curran, J. M., & Buckleton, J. S. (2008). Effectiveness of familial searches. *Science & Justice*, *48*, 164–167.

Epstein, J. (2009). "Genetic surveillance" – the bogeyman response to familial DNA investigations. *University of Illinois Journal of Law, Technology, and Policy*, *1*, 141–173.

Flimmers, R., et al. (2008). STR-profiling for the differentiation between related and unrelated individuals in cases of citizen rights. *Forensic Science International: Genetics Supplement Series*, *1*, 510–513.

Forensic Science Service. (2008). Craig Harman – family DNA link offers crime breakthrough. Available at <http://www.forensic.gov.uk/html/media/case-studies/f-39.html>.

Fu, J., et al. (2007). Considerations for the interpretation of STR results in cases of questioned half-sibship. *Transfusion*, *47*, 515–519.

Gabel, J. D. (2010). Probable cause from probable bonds: A genetic tattle tale based on familial DNA. *Hastings Women's Law Journal*, *21*, 3–57.

Ge, J., et al. (2010). Comparing DNA based familial searching policies. *Proceedings of the twenty first international symposium on human identification*. Available at <http://www.promega.com/geneticidproc/ussymp21proc/oralpresentations/Ge.pdf>.

Gershaw, C. J., et al. (2011). Forensic utilization of familial searches in DNA databases. *Forensic Science International: Genetics*, *5*, 16–20.

Greely, H. T., et al. (2006). Family ties: The use of DNA offender databases to catch offenders' kin. *Journal of Law, Medicine & Ethics*, *34*, 248–252.

Grimm, D. J. (2007). The demographics of genetic surveillance: Familial DNA testing and the Hispanic community. *Columbia Law Review*, *107*, 1164–1194.

Hicks, T., et al. (2010). Use of DNA profiles for investigation using a simulated national DNA database: Part II. Statistical and ethical considerations on familial searching. *Forensic Science International: Genetics*, *4*, 316–322.

Kaye, D. H. (2010). "Familial searching": Ten questions and answers. Available at <http://www.personal.psu.edu/dhk3/blogs/DoubleHelixLaw/blog/>.

Lewis, K. E. (2009). Genomic approaches to forensic DNA analysis. PhD dissertation. University of Washington, Department of Genome Sciences.

Liu, H.-M., et al. (2007). Y-chromosome short tandem repeats analysis to complement paternal lineage study: A single institutional experience in Taiwan. *Transfusion*, *47*, 918–926.

Miller, G. (2010). Familial DNA testing scores a win in serial killer case. *Science*, *329*, 262.

Myers, S. P., et al. (2010). Searching for first-degree familial relationships in California's offender DNA database: Validation of a likelihood ratio-based approach. *Forensic Science International: Genetics* (in press). doi:10.1016/j.fsigen.2010.10.010.

Murphy, E. (2010). Relative doubt: Familial searches of DNA databases. *Michigan Law Review*, *109*, 291–348.
O'Connor, K. L., et al. (2010). Evaluating the effect of additional forensic loci on likelihood ratio values for complex kinship analysis. *Proceedings of the twenty first international symposium on human identification*. Available at <http://www.promega.com/geneticidproc/ussymp21proc/oralpresentations/OConnor.pdf>.
Presciuttini, S., et al. (2003). Allele sharing in first-degree and unrelated pairs of individuals in the GeFI AmpFlSTR Profiler Plus database. *Forensic Science International*, *131*, 85–89.
Reid, T. M., et al. (2004). Specificity of sibship determination using the ABI Identifiler multiplex system. *Journal of Forensic Sciences*, *49*, 1262–1264.
Reid, T. M., et al. (2008). Use of sibling pairs to determine the familial searching efficiency of forensic databases. *Forensic Science International: Genetics*, *2*(4), 340–342.
Rozen, J. (2009, March 17). Genetic surveillance for all. *Slate Magazine*. Available at <http://www.slate.com/id/2213958/>.
Sjerps, M., & Kloostermann, A. D. (1999). On the consequences of DNA profile mismatches for close relatives of an excluded suspect. *International Journal of Legal Medicine*, *112*, 176–180.
Steinberger, E., & Sims, G. (2008). Finding criminals through the DNA of their relatives—familial searching of the California offender DNA database. *Prosecutor's Brief*, *31*, 28–32.
Suter, S. M. (2010). All in the family: Privacy and DNA familial searching. *Harvard Journal of Law & Technology*, *23*, 309–399.
SWGDAM (2009). SWGDAM recommendations to the FBI Director on the "Interim plan for the release of information in the event of a 'partial match' at NDIS". *Forensic Science Communications*, *11*(4). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/oct2009/standard_guidlines/swgdam.html>.
Willing, R. (2005, June 8). Suspects get snared by a relative's DNA. *USA Today*. Available at <http://www.usatoday.com/news/nation/2005-06-07-dna-cover_x.htm>.

Grim Sleeper Case and California Familial Search Policy

ACLU (2010). "Grim Sleeper" case doesn't justify expanding the reach of DNA databases. <http://www.aclu-sc.org/news_stories/view/102863/>.
Dolan, M., & Felch, J. (2008, November 25). Tracing a suspect through a relative. *Los Angeles Times*. Available at <http://articles.latimes.com/2008/nov/25/local/me-familial25>.
Dolan, M., et al. (2010, July 8). DNA leads to arrest in Grim Sleeper killings. *Los Angeles Times*. Available at <http://articles.latimes.com/2010/jul/08/local/la-me-grim-sleeper-20100708>.
Editorial (2010, July 13). A yellow light to DNA searches. *New York Times*. Available at <http://www.nytimes.com/2010/07/13/opinion/13tue1.html>.
Miller, G. (2010). Familial DNA testing scores a win in serial killer case. *Science*, *329*, 262.
Myers, S. P., et al. (2010). Searching for first-degree familial relationships in California's offender DNA database: Validation of a likelihood ratio-based approach. *Forensic Science International: Genetics* (in press). doi:10.1016/j.fsigen.2010.10.010.
Pelisek, C. (2008, August 28). Grim Sleeper returns: He's murdering Angelenos, as cops hunt his DNA. *LA Weekly*. Available at <http://www.laweekly.com/2008-08-28/news/grim-sleeper/>.
Pelisek, C. (2008, August 28). Grim Sleeper's victims. *LA Weekly*. Available at <http://www.laweekly.com/2008-08-28/news/eleven-lives-stolen-and-one-lucky-survivor/>.
Sims, G., et al. (2008). The DNA partial match and familial search policy of the California Department of Justice. *Proceedings of the nineteenth international symposium on human identification.* Available at <http://www.promega.com/geneticidproc/ussymp19proc/oralpresentations/Sims.pdf>.
Steinhauer, J. (2010, July 8). "Grim Sleeper" arrest fans debate on DNA use. *New York Times*. Available at <http://www.nytimes.com/2010/07/09/us/09sleeper.html>.